DO NOW — RECALL

# Data mining and visualisation

Name: _____ Class: _____ Date: _____

**1. What is data mining?**

**2. What is an outlier?**

**3. What type of graph shows two numeric variables?**

**4. C# int[] scores={2,4,6,8}. Double elements at indexes 0 and 2; give the final values and total.**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

**1. Model answer**

Identifying meaningful patterns in datasets.

**2. Model answer**

Value unusually distant from the rest.

**3. Model answer**

Scatter graph.

**4. Model answer**

[4,4,12,8], total 28.

**By the end, you can…**

• Identify the question

• Choose a suitable chart

• Interpret responsibly

**Vocabulary — know these terms**

data mining — Extracting patterns from datasets.

visualisation — Graphical representation of data.

correlation — Statistical association, not proof of cause.

outlier — Observation unusually far from others.

NEW LEARNING

# Read, annotate and explain

### Identify the question

Data mining searches for useful patterns in a set of records; decide what question the analysis should answer.

### Choose a suitable chart

Scatter charts show relationships between two numeric variables; bars compare categories; histograms show numeric distributions.

### Interpret responsibly

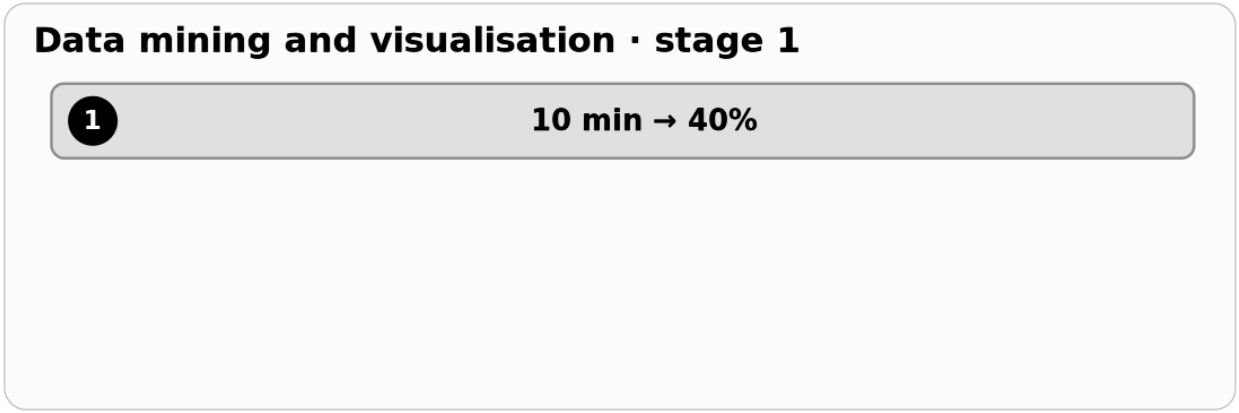
An association does not prove that one variable causes another. Consider sample size, bias, missing data and outliers.

### Explain one step in your own words, using the correct technical terms.

# New Learning · original lesson diagrams

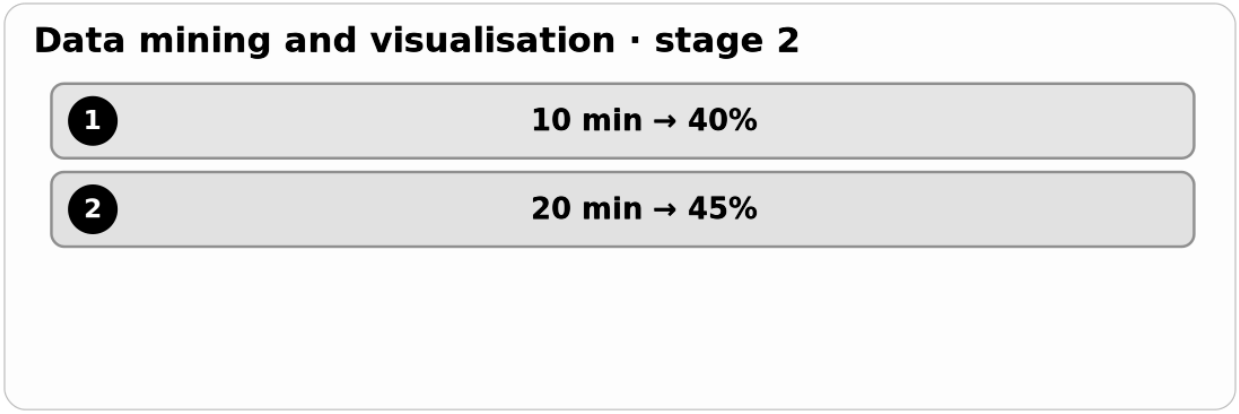
Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




Data mining and visualisation · stage 1; 10 min → 40%

### New Learning




Data mining and visualisation · stage 2; 10 min → 40%; 20 min → 45%

# New Learning · original lesson diagrams

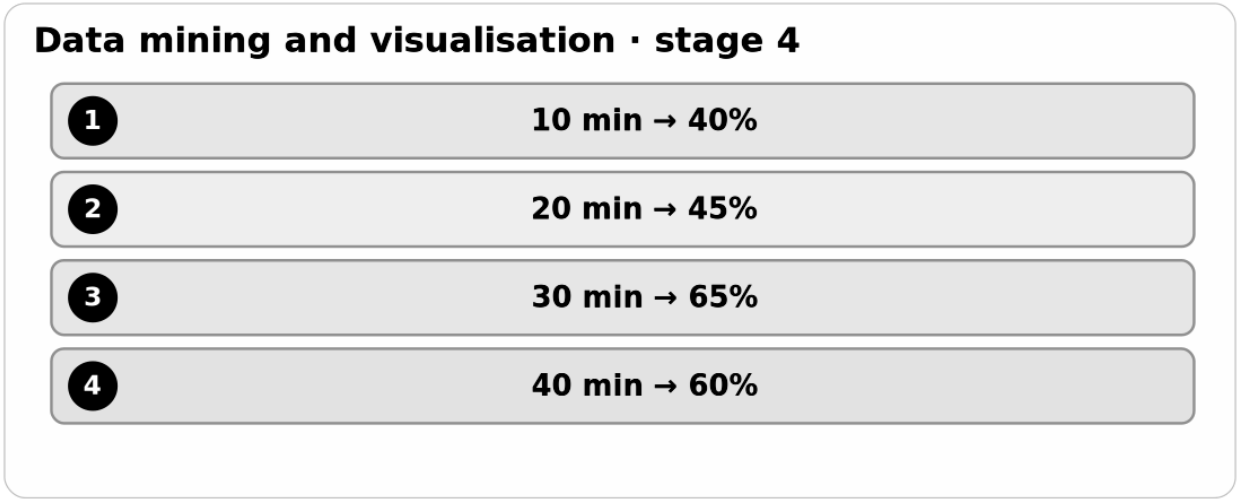
Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




Data mining and visualisation · stage 3; 10 min → 40%; 20 min → 45%; 30 min → 65%

### New Learning

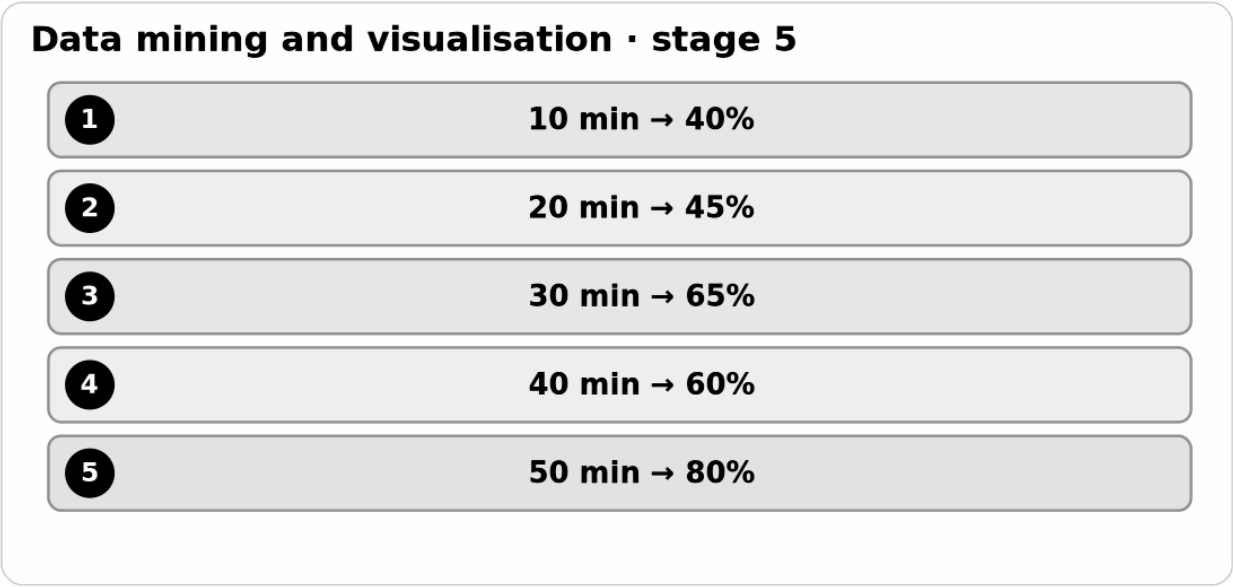



Data mining and visualisation · stage 4; 10 min → 40%; 20 min → 45%; 30 min → 65%; 40 min → 60%

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




Data mining and visualisation · stage 5; 10 min → 40%; 20 min → 45%; 30 min → 65%; 40 min → 60%; 50 min → 80%

I DO — WORKED EXAMPLE

# Follow the worked method

**For observations (1,2), (2,4), (3,6), describe the association without claiming causation.**

### Predict first

As x increases, y increases in this small sample, indicating positive association.

### Explain the method

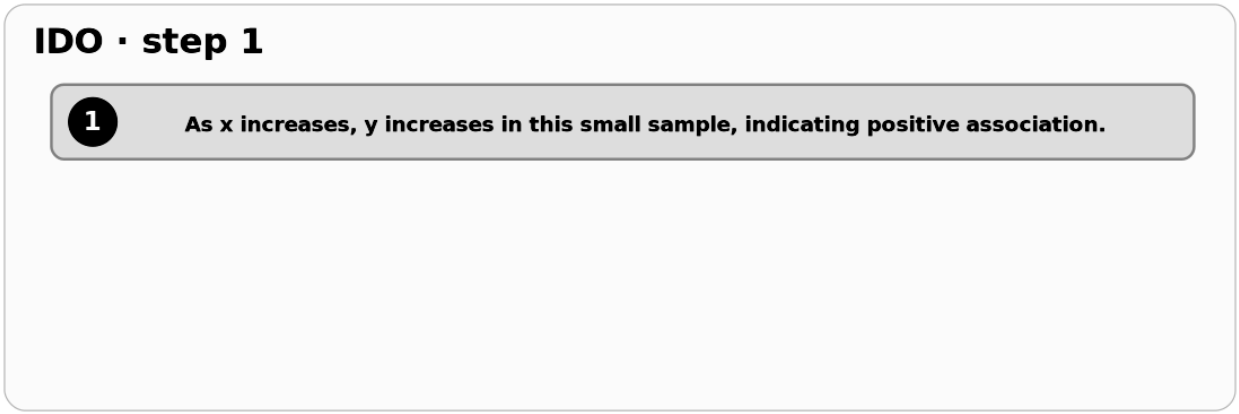
No causal claim is justified without additional evidence or experimental design.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### For observations (1,2), (2,4), (3,6), describe the association without claiming causation.

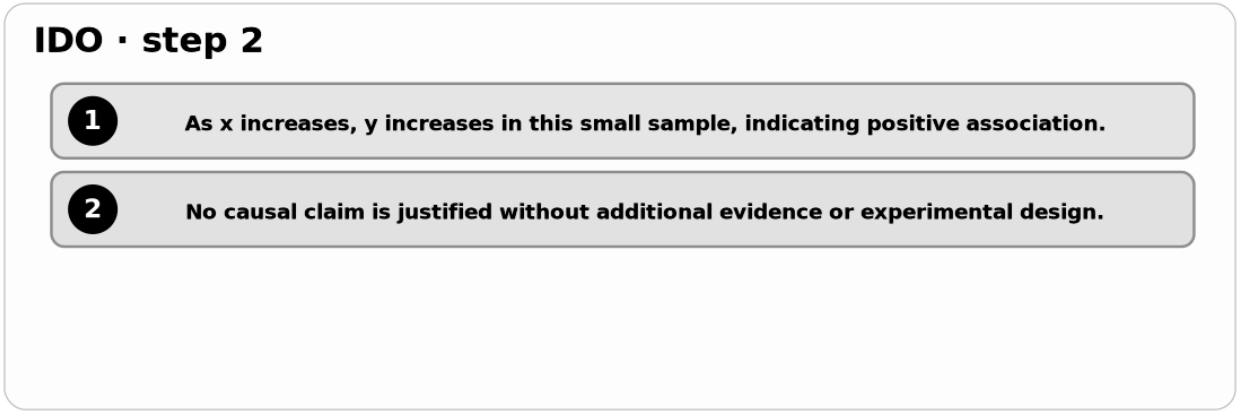
As x increases, y increases in this small sample, indicating positive association.




IDO · step 1; As x increases, y increases in this small sample, indicating positive association.

### For observations (1,2), (2,4), (3,6), describe the association without claiming causation.

As x increases, y increases in this small sample, indicating positive association.




IDO · step 2; As x increases, y increases in this small sample, indicating positive association.; No causal claim is justified without additional evidence or experimental design.

WE DO — GUIDED PRACTICE

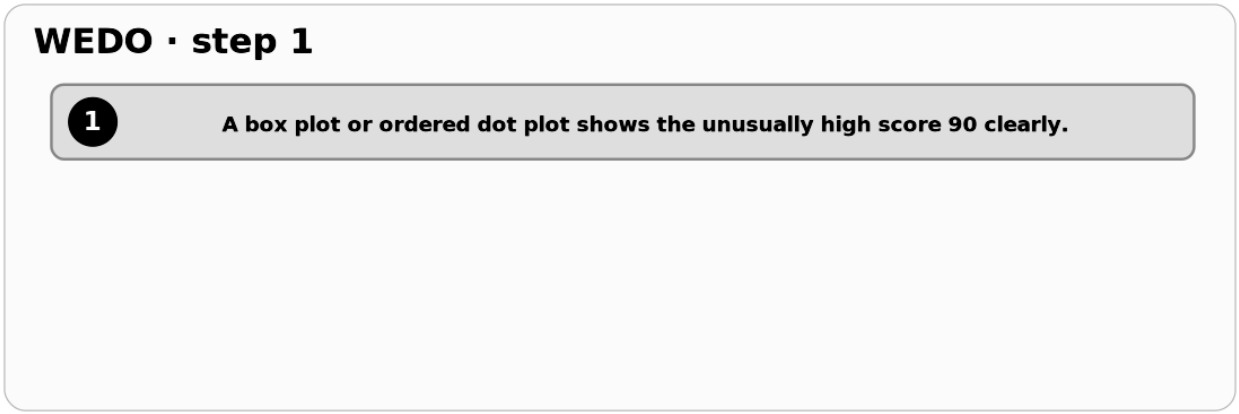
# Try → Discuss → Improve

# We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### For scores [40,42,45,90], select a suitable visual and explain a limitation of the mean.

A box plot or ordered dot plot shows the unusually high score 90 clearly.




WEDO · step 1; A box plot or ordered dot plot shows the unusually high score 90 clearly.

# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

### Guided 1: Predict first A box plot or ordered dot plot shows the unusually high score 90 clearly.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

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YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Choose and justify a chart for (a) scores by category (b) minutes versus score (c) frequency by score interval. [6]

### Question 2

For [40,42,45,90], calculate mean, median and identify one misleading interpretation. [6]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Evaluate a school's plan to mine pupil browsing records to predict examination outcomes. [6]

**EXIT TICKET / REFLECTION**

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**If two variables are correlated, the first must cause the second.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

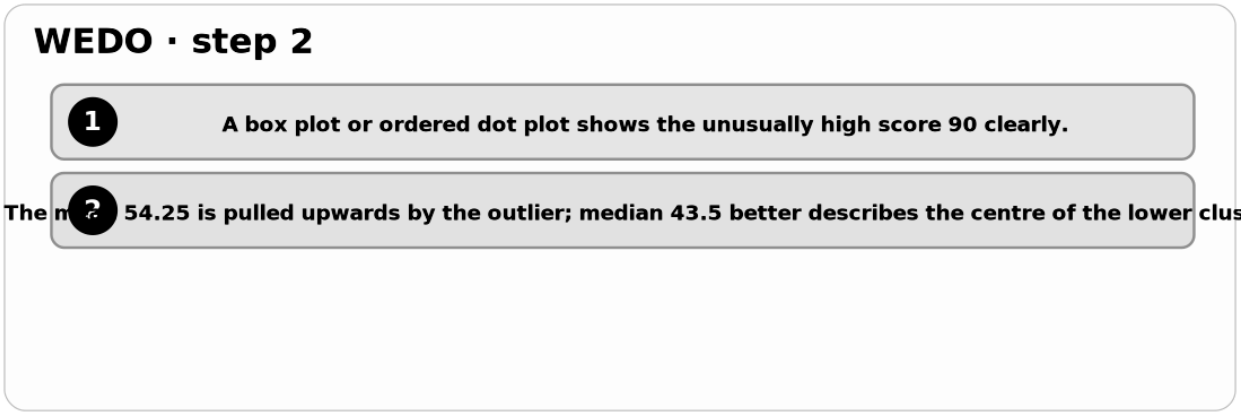
Next step: revisit the model or a relevant revision-guide section if needed.

# We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### For scores [40,42,45,90], select a suitable visual and explain a limitation of the mean.

A box plot or ordered dot plot shows the unusually high score 90 clearly.




WEDO · step 2; A box plot or ordered dot plot shows the unusually high score 90 clearly.; The mean 54.25 is pulled upwards by the outlier; median 43.5 better describes the centre of the lower cluster.

# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

### Guided 1 — original CHECK

Explain the method

The mean 54.25 is pulled upwards by the outlier; median 43.5 better describes the centre of the lower cluster.

IMPROVE: Explain one correction to your own reasoning.